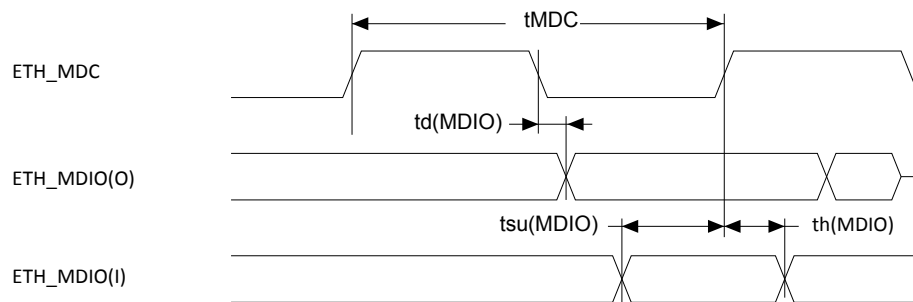


Figure 42. SMI timing diagram


DT31384V1

5.3.30 JTAG/SWD interface characteristics

Unless otherwise specified, the parameters given in [Table 75](#) and [Table 76](#) are derived from tests performed under the ambient temperature, f_{HCLKx} frequency and V_{DD} supply voltage conditions summarized in [Table 18](#), with the following configuration:

- Output speed set to OSPEEDRy[1:0] = 10
- Capacitive load $C_L = 30$ pF
- Measurement points done at $0.5 \times V_{DD}$ level

Refer to [Table 48. I/O static characteristics](#) for more details on the input/output characteristics.

Table 75. JTAG characteristics

Evaluated by characterization - Not tested in production.

Symbol	Parameter	Min	Typ	Max	Unit
F_{TCK}	TCK clock frequency	-	-	34	MHz
$t_{su}(TMS)$	TMS input setup time	2	-	-	ns
$t_{h}(TMS)$	TMS input hold time	0.5	-	-	
$t_{su}(TDI)$	TDI input setup time	1.5	-	-	
$t_{h}(TDI)$	TDI input hold time	0.5	-	-	
$t_{ov}(TDO)$	TDO output valid time	-	11	14.5	
$t_{oh}(TDO)$	TDO output hold time	7.5	-	-	